

Employment Effects of Canadian Minimum Wage Increases, 2023–2025: Evidence from Industry-Level Dynamic Difference-in-Differences

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Abstract

Following the post-pandemic recovery of the labor market, Canadian provinces have implemented frequent adjustments to minimum wage levels since 2023. This institutional environment provides a new empirical setting for identifying the employment effects of minimum wage policies.

Using variation in the timing and magnitude of provincial minimum wage adjustments, we find that employment responses differ substantially across industries and between the short run and the medium run. In the short run, employment does not exhibit large systematic changes on average, although the evidence suggests the presence of adjustment pressures following minimum wage increases. In the medium run, only a small number of industries display statistically significant negative employment effects, while most industries show effects that are statistically indistinguishable from zero.

Overall, the results suggest that recent minimum wage increases in Canada have not generated broad-based employment losses, though negative effects emerge in a limited subset of industries. Further evidence from longer horizons will be essential for assessing whether these patterns persist as minimum wage policies continue to evolve in Canada's post-pandemic labor market.

Keywords: minimum wage; employment; industry heterogeneity; difference-in-differences

1 Introduction

This paper studies the employment effects of recent minimum wage increases in Canada following the post-2023 policy adjustments. We combine a structural framework with imperfect competition in labor and product markets with industry-level evidence to interpret heterogeneous employment responses to minimum wage changes. At the same time, the high frequency and relatively low intensity of stepwise minimum wage increases pose challenges for traditional empirical approaches that focus on average treatment effects. The central objective of this paper is therefore to characterize heterogeneous employment responses to minimum wage policies across industries and over different time horizons, guided by explicit theoretical mechanisms.

Section 2 reviews and synthesizes the core theoretical and empirical developments in the minimum wage literature. We first discuss the classical result that minimum wages necessarily reduce employment under perfectly competitive labor markets and highlight the limitations of its underlying assumptions in low-wage labor markets. The section then introduces minimum wage theories under unilateral and bilateral imperfect competition, emphasizing monopsonistic labor demand and demand-side feedback effects in product markets. A key insight emerging from this discussion is that the employment effects of minimum wages are inherently context-dependent and industry-specific, implying that historical estimates may offer limited guidance for newly enacted policy regimes.

Section 3 describes the data sources and sample construction. The empirical analysis relies on a newly constructed province–industry–month panel dataset for Canada that combines labor market information from Statistics Canada with statutory minimum wage data from Employment and Social Development Canada. The section details the construction of consistent employment rate measures at the province–industry level, the harmonization of industry classifications across data sources, and the mapping of minimum wage changes into a monthly policy series. The final sample covers ten Canadian provinces from January 2023 to September 2025, subject to data availability by industry.

Section 4 develops a structural theoretical framework that incorporates imperfect competition

in both labor and product markets. The framework illustrates how minimum wages can generate a non-monotonic relationship with employment by simultaneously relaxing wage-setting distortions on the labor demand side and strengthening income-driven demand feedback on the product market side. This section yields clear, testable predictions regarding heterogeneous employment responses across industries¹, which guide the subsequent empirical analysis.

Section 5 characterizes the model’s equilibrium outcomes numerically under alternative parameter configurations, including variations in labor supply elasticity, capital–labor substitutability, capital intensity, and the income sensitivity of product demand. By mapping abstract theoretical mechanisms to observable industry characteristics, this section derives a set of industry-level predictions relevant for the Canadian context and provides a structural interpretation for the empirical results.

Section 6 describes the empirical research design. Exploiting monthly variation in the timing and magnitude of provincial minimum wage adjustments since 2023, the empirical framework distinguishes between short-run adjustment dynamics and medium-run cumulative employment effects. This section highlights how, in the presence of multiple and staggered policy changes, failing to separate these time dimensions may confound historical policy exposure with contemporaneous adjustment effects, thereby obscuring the dynamic employment responses to minimum wage increases.

Section 7 presents the main empirical findings. The results reveal substantial heterogeneity in employment responses across industries. In the short run, several industries exhibit statistically significant dynamic adjustments around minimum wage increases, though there is no evidence of a uniform directional employment shock. In the medium run, a limited set of industries—including accommodation and food services, transportation and warehousing, and professional services—experience statistically significant negative employment responses, while the majority of industries, including low-wage sectors, display effects that are statistically indistinguishable from

¹Industry-level analysis is particularly informative in this setting because minimum wage exposure, production technology, and demand conditions vary systematically across sectors, while worker-level data may confound industry adjustment with worker reallocation.

zero. The section further relates these empirical patterns to the theoretical predictions developed in Sections 3 and 4.

Section 8 conducts a series of robustness checks and diagnostic exercises, including heterogeneity by event spacing, placebo timing tests, permutation-based inference, and an examination of cross-province spillover effects. These analyses demonstrate that the main findings are not mechanically driven by policy timing, macroeconomic co-movement, or spatial correlation, but instead reflect genuine adjustment processes shaped by industry structure.

Section 9 concludes by discussing the policy implications of the findings. The paper emphasizes that, in the post-pandemic institutional environment, recent minimum wage increases in Canada have not generated widespread employment losses, but have produced non-negligible negative effects in specific industries and time horizons. The conclusion also highlights the need for longer time-series data to assess whether these medium-run adjustments evolve into more persistent structural effects.

2 Literature Review

2.1 Classical Competitive Models of the Minimum Wage

Early theoretical analyses of minimum wage policies are rooted in the perfectly competitive labor market framework. In the canonical model formalized by Stigler (1946), firms are wage takers and employment is determined by the intersection of labor demand and labor supply. A binding minimum wage raises the cost of labor above its marginal productivity for low-skilled workers, leading firms to reduce employment. Within this framework, minimum wage increases unambiguously generate employment losses, particularly among young and low-skilled workers.²

This prediction shaped much of the early empirical literature, which generally relied on time-series or cross-sectional comparisons and often found negative employment effects. However, as

²The perfectly competitive model serves primarily as a theoretical benchmark. Its predictions need not hold once search frictions, wage-setting power, or product market imperfections are introduced.

later research emphasized, the competitive model rests on strong assumptions—perfect worker mobility, no search frictions, and the absence of firm-level wage-setting power—that are increasingly viewed as empirically implausible, especially in low-wage labor markets characterized by high turnover, local monopsony power, and limited outside options.

These limitations motivated the development of alternative theoretical frameworks that relax the assumption of perfect competition in labor markets.

2.2 Monopsonistic Labor Markets and Wage-Setting Power

A central theoretical contribution to the modern minimum wage literature is the monopsony model of labor markets, formalized and extended by Manning (2003). In this framework, firms face upward-sloping labor supply curves due to search frictions, job differentiation, geographic immobility, and heterogeneous worker preferences. As a result, firms possess wage-setting power and internalize the effect of their hiring decisions on wages.³

In monopsonistic equilibrium, wages are set below the marginal revenue product of labor, leading to inefficiently low employment. A moderate increase in the minimum wage can therefore reduce the wedge between wages and productivity, raising both wages and employment. Importantly, the employment response to a minimum wage is non-monotonic: employment increases for minimum wages below the firm's effective marginal labor cost, but declines once the minimum wage becomes sufficiently high.

This insight fundamentally altered the theoretical debate by demonstrating that minimum wage increases need not reduce employment when labor markets are imperfectly competitive. Subsequent theoretical work has refined and generalized this insight. Bhaskar and To (1999) analyze minimum wages in a setting with imperfect competition on both sides of the market, showing that minimum wages can enhance allocative efficiency under monopsonistic distortions. Similarly, Manning (2003) emphasizes that monopsony power can arise even in markets with many firms, as

³Empirically, monopsony power may arise even in large labor markets when workers face frictions such as job-specific amenities, switching costs, or geographic constraints.

long as workers face frictions in job search and mobility.

These models provide a clear rationale for heterogeneous employment effects across industries and regions, depending on the degree of labor market power and the elasticity of labor supply faced by firms.

2.3 Imperfect Competition in Product Markets and Demand-Side Channels

While much of the minimum wage literature focuses on labor market distortions, a growing body of work emphasizes the importance of product market structure and demand-side channels. In many low-wage industries—such as retail, food services, and personal services—firms operate in imperfectly competitive product markets and sell primarily to local consumers. In such settings, changes in household income can feed back into product demand.

Theoretical models incorporating imperfect competition in product markets suggest that higher wages can raise local demand by increasing workers' purchasing power, thereby expanding output and labor demand. This mechanism is particularly relevant for non-tradable, locally consumed goods and services, where higher local income translates directly into higher demand. By contrast, in tradable or capital-intensive sectors, demand is less sensitive to local income, and firms can more easily substitute capital for labor.

Although this income-driven demand channel is often discussed informally, it has received relatively limited formal treatment in structural models of minimum wages. Some recent contributions emphasize that ignoring product market feedback can lead to incomplete or misleading conclusions regarding employment effects. By explicitly modeling household demand and price formation, these frameworks highlight how minimum wage policies can affect both marginal revenue and marginal cost simultaneously.

The interaction between labor market monopsony and product market imperfections is therefore central to understanding heterogeneous employment responses across industries.

2.4 Empirical Evidence: From Average Effects to Heterogeneity

Empirically, the modern minimum wage literature has evolved from estimating average employment effects toward identifying heterogeneous responses across workers, industries, and local labor markets.

Card and Krueger (1994) provided an influential early challenge to the competitive model by exploiting a natural experiment in the fast-food industry. Using a difference-in-differences approach, they found no evidence of employment losses following a minimum wage increase in New Jersey relative to Pennsylvania. This study sparked extensive debate and prompted a re-examination of empirical methods and identifying assumptions.

Subsequent studies using improved data and methods have produced more nuanced results. Brochu and Green (2013), focusing on Canadian minimum wage increases, find small average employment effects but substantial heterogeneity across demographic groups and local labor market conditions. Their results suggest that minimum wage impacts depend critically on institutional context and labor market structure.

Clemens and Strain (2017) examine recent U.S. minimum wage increases using difference-in-differences and triple-differences designs, focusing on low-skilled and young workers. They find that employment among these groups grew more slowly in states that raised minimum wages, suggesting modest short-run negative effects. However, they also emphasize that these effects are far from uniform and vary across sectors and regions.

More recent empirical work increasingly emphasizes industry-level heterogeneity. Studies show that employment effects are often concentrated in capital-intensive or tradable sectors, while low-wage, labor-intensive service industries tend to exhibit neutral or small effects. These findings are broadly consistent with monopsony-based theories and demand-side mechanisms, but existing empirical evidence remains fragmented.

2.5 Linking Theory and Evidence: Industry Heterogeneity and Mechanisms

Despite significant progress, a gap remains between theoretical models emphasizing market imperfections and empirical studies documenting heterogeneous employment responses. Many empirical analyses rely on reduced-form designs that identify average treatment effects but do not directly link observed heterogeneity to underlying structural features such as labor supply elasticity, production technology, or demand sensitivity.

At the same time, existing theoretical models often abstract from explicit product demand feedback or treat prices as exogenous. As a result, they cannot fully capture the interaction between wage-setting power, household income, and product market equilibrium that appears central in low-wage, locally consumed industries.

This paper contributes to the literature by developing a unified framework that integrates monopsonistic competition in labor markets with imperfect competition and income-driven demand in product markets. By explicitly modeling household labor supply, household consumption decisions, and firm optimization, the framework generates clear, testable predictions regarding when and where minimum wage increases can raise employment and output.

In particular, the model predicts that minimum wages are most likely to generate positive or neutral employment effects in industries characterized by strong monopsony power, labor-intensive production, limited capital–labor substitution, and demand that is sensitive to local income. Conversely, employment losses are more likely in capital-intensive, tradable, or high-skill sectors where these conditions do not hold.

These predictions motivate the empirical analysis that follows, which tests for systematic heterogeneity in minimum wage effects across Canadian industries using difference-in-differences and triple-differences designs.

3 Data Sources and Variable Construction

This paper constructs a province–industry–month panel dataset for Canada to analyze the employment effects of minimum wage policies. The analysis combines labour market outcome variables from Statistics Canada with statutory minimum wage information from Employment and Social Development Canada (ESDC). To ensure consistency across data sources in terms of time coverage, geographic units, and industry classification, the final sample is restricted to ten Canadian provinces.

3.1 Labour Market Outcomes

This study constructs a province–industry–month panel dataset for Canada to examine the labour market effects of minimum wage adjustments. The main outcome variable is the employment rate. Because employment rates are not directly available at the province–industry–month level, they are constructed using data from Statistics Canada’s Labour Force Survey (LFS). Specifically, employment and labour force measures are obtained from Statistics Canada Table 14-10-0022-01 (PID: 1410002201), which reports monthly labour market characteristics by province and industry. The employment rate is defined as the ratio of employment to the labour force at the province–industry–month level.⁴⁵

In addition, average hourly wage data are obtained from Statistics Canada Table 14-10-0063-01 (PID: 1410006301), which provides monthly wage information for employees by province and industry. The employment rate and average wage variables jointly capture labour market responses to minimum wage adjustments.

⁴Employment rates at the province–industry–month level are not directly reported by Statistics Canada. We construct them using employment and labour force counts from Table 14-10-0022-01. Observations flagged as confidential or suppressed (e.g., “x” or “.”) under the Statistics Act are treated as missing.

⁵Employment rates are used rather than employment levels to facilitate comparison across industries of different sizes and to abstract from long-run demographic trends.

3.2 Minimum Wage Data

Statutory minimum wage data are obtained from the official Minimum Wage Database maintained by Employment and Social Development Canada (ESDC), a federal government department responsible for labour market policies and employment standards. The database reports legally mandated minimum wage rates by province along with their effective dates. Because minimum wage changes occur discretely and at different times across provinces, statutory rates are converted into a monthly series by assigning each minimum wage level to all months during which it is legally in force.⁶ The database is publicly available at <https://minwage-salairemin.service.canada.ca/en/since1965.html>.

Although the labour market outcome variables primarily cover the period from January 2023 to September 2025, minimum wage information from 2022 is also incorporated to construct baseline levels and lagged policy variables used in the empirical analysis.

3.3 Data Processing and Sample Coverage

Several data processing steps are implemented to ensure internal consistency across data sources. Employment, labour force, and wage data differ in the level of industry disaggregation across tables; therefore, the analysis is restricted to industries that are jointly observed and consistently defined across the relevant Statistics Canada tables. Industry labels are standardized to remove minor naming differences, and all variables are aligned at the province–industry–month level. Observations flagged as confidential or suppressed in the original data are treated as missing.⁷

The final dataset covers ten Canadian provinces and spans the period from 2023 to 2025, providing a consistent panel structure for the empirical analysis.

⁶Monthly aggregation may smooth very short-lived adjustments occurring within the implementation month. The analysis therefore focuses on dynamic responses over several months rather than immediate discontinuities.

⁷Restricting attention to consistently defined industries reduces measurement error but limits the analysis of more disaggregated occupational or firm-level adjustment.

3.4 Macroeconomic Controls (Robustness)

To control for contemporaneous macroeconomic conditions that may jointly affect employment outcomes and minimum wage adjustments, the empirical analysis introduces a set of province-level macroeconomic variables in robustness specifications. Provincial population data are obtained from Statistics Canada Table 17-10-0009-01. Monthly provincial consumer price index (CPI) data are drawn from Statistics Canada Table 18-10-0004-01 (formerly CANSIM 326-0020), which reports CPI by province. Monthly provincial retail sales data are obtained from Statistics Canada Table 20-10-0056-01, which reports seasonally adjusted retail trade sales by province and territory. The use of seasonally adjusted retail sales helps abstract from regular seasonal fluctuations and capture underlying short-run macroeconomic demand conditions.

All macroeconomic variables are organized at the province–month level and merged with the province–industry–month panel. Variables measured in levels are typically transformed into logarithms in the regression specifications. These macroeconomic controls are included only in robustness exercises and are excluded from the baseline specifications, as conditioning on contemporaneous macro variables may absorb part of the policy-induced adjustment process and mechanically attenuate the estimated employment effects of minimum wage changes.

4 Theoretical Model

We consider an economy characterized by imperfect competition in both labor and product markets. Firms face downward-sloping demand curves in the product market and upward-sloping labor supply curves in the labor market due to search frictions, geographic immobility, and workers’ outside options. Even when many firms are present, each firm internalizes the effect of its own hiring decisions on wages, implying wage-setting power.

4.1 Firms and Production Technology

A representative firm produces output using labor L and capital K . Capital is fixed in the short run. Output is produced using a constant-returns-to-scale CES production technology:

$$Q = A [\delta K^{\rho_p} + (1 - \delta)L^{\rho_p}]^{1/\rho_p}, \quad (1)$$

where $\delta \in (0, 1)$ governs capital intensity and ρ_p determines the elasticity of substitution between capital and labor.

Firms sell output in an imperfectly competitive product market and face an inverse demand curve $P(Q)$ with $P'(Q) < 0$. Firm profits are given by:

$$\pi(Q) = P(Q)Q - w(L(Q))L(Q) - rK. \quad (2)$$

4.2 Households, Labor Supply, and Wage Determination

Households derive utility from consumption and leisure. Each household is endowed with one unit of time, which can be allocated between labor supply $L \in [0, 1]$ and leisure $1 - L$. In addition to labor income, households may receive non-labor income from alternative sources such as transfers, family support, or home production.

Household preferences are represented by:

$$U(C, 1 - L) = \ln(C) + \gamma \ln(1 - L), \quad (3)$$

where C denotes consumption and $\gamma > 0$ governs the relative preference for leisure.

Consumption is given by:

$$C = wL + Y_0, \quad (4)$$

where $Y_0 \geq 0$ denotes non-labor income. Given the wage w , households choose labor supply to

maximize utility:

$$\max_{L \in [0,1]} \ln(wL + Y_0) + \gamma \ln(1 - L). \quad (5)$$

The first-order condition implies the labor supply function:

$$L^s(w) = \frac{1}{1 + \gamma} - \frac{\gamma Y_0}{(1 + \gamma)w}, \quad (6)$$

which is increasing in w and bounded above by $1/(1 + \gamma)$. The implied inverse labor supply curve is:

$$w(L) = \frac{\gamma Y_0}{1 - (1 + \gamma)L}. \quad (7)$$

4.3 Households, Product Demand, and Price Formation

In addition to supplying labor, households act as consumers in the product market. Household income consists of labor earnings and non-labor income and is given by:

$$I = wL + Y_0. \quad (8)$$

Households allocate this income between the consumption of a differentiated good local good q and numeraire good c .

Good q is interpreted as a locally consumed, non-tradable good or service, such as retail services, food services, or other local activities. Its price is determined within the local market and is potentially affected by changes in household income. By contrast, the numeraire good c represents a composite of other consumption goods whose price is normalized to one and taken as exogenous. This good may be interpreted as a nationally or globally traded bundle of goods and serves to close the household budget constraint and generate income effects in product demand.

Household preferences over consumption are represented by the utility function:

$$u(q, c) = aq - \frac{b}{2}q^2 + \frac{c^{1-\rho_d}}{1-\rho_d}, \quad \rho_d > 0. \quad (9)$$

The first two terms capture diminishing marginal utility from consumption of the differentiated good, while the last term captures utility from the numeraire good. The parameter ρ_d governs the strength of income effects in product demand: when ρ_d is larger, changes in household income translate more strongly into changes in the willingness to pay for the differentiated good.

Households face the budget constraint:

$$c + Pq = I, \quad (10)$$

where P denotes the price of the differentiated good. Given prices and income, households choose (q, c) to maximize utility.

The first-order conditions for an interior solution are:

$$a - bq = \lambda P, \quad (11)$$

$$c^{-\rho_d} = \lambda, \quad (12)$$

where λ denotes the marginal utility of income. Eliminating λ yields the household inverse demand condition:

$$P = (a - bq)c^{\rho_d}. \quad (13)$$

This expression has a clear economic interpretation. For a given level of consumption of the differentiated good, households are willing to pay a higher price when (i) the marginal utility of q is high and (ii) overall consumption of the numeraire good is high, reflecting higher household income. As a result, the willingness to pay for the differentiated good depends explicitly on household income.

Substituting the budget constraint into the inverse demand condition, $c = I - Pq$, and imposing market clearing in the differentiated good market ($q = Q$), the equilibrium price must satisfy:

$$P(Q, I) = (a - bQ)(I - P(Q, I)Q)^{\rho_d}. \quad (14)$$

This equation defines the inverse demand curve faced by firms in the product market. Importantly, prices are not directly chosen by households. Instead, household optimization pins down the willingness to pay for each level of aggregate output, while firms take this inverse demand relationship as a constraint when choosing output. Prices are therefore formed in equilibrium through the interaction of household demand and firm supply.

Because household income depends on wages and employment, which are themselves endogenously determined in the labor market, the inverse demand curve shifts with labor market conditions. In particular, increases in wages or employment raise household income and shift the inverse demand curve outward, increasing the equilibrium price and quantity of the differentiated good. This income-driven demand channel links labor market outcomes to product market equilibrium and plays a central role in the transmission of minimum wage policies in the model.

4.4 Equilibrium and the Effects of a Minimum Wage

In the absence of a minimum wage, firms choose output to maximize profits while internalizing the wage response implied by the upward-sloping labor supply curve. The firm's optimal output choice satisfies the first-order condition:

$$P(Q) + QP'(Q) - [w(L) + Lw'(L)] \frac{dL}{dQ} = 0. \quad (15)$$

The term $w(L) + Lw'(L)$ represents the effective marginal labor cost faced by the firm. In addition to paying the wage required to attract marginal workers, firms must account for the increase in wages paid to inframarginal workers induced by higher employment. This wage-pushing effect is the central distortion generated by monopsonistic competition and leads firms to restrict employment and output relative to the competitive benchmark.

Consider now the introduction of a minimum wage $W_{\min}$. If the minimum wage lies below the equilibrium wage, it is non-binding and does not affect firm behavior. When $W_{\min}$ exceeds the equilibrium wage, wages become fixed at $W_{\min}$ up to the labor supply constraint. Firm profits are

then given by:

$$\pi(Q) = P(Q)Q - W_{\min}L(Q) - rK. \quad (16)$$

The corresponding first-order condition becomes:

$$P(Q) + QP'(Q) - W_{\min} \frac{dL}{dQ} = 0. \quad (17)$$

Relative to the no-minimum-wage case, the wage-pushing term $Lw'(L)$ disappears from the firm's optimization problem. As long as the minimum wage remains below the original effective marginal labor cost, fixing the wage at $W_{\min}$ reduces the marginal cost of expanding employment. This labor market channel creates a range of minimum wages—referred to as the *effective minimum wage interval*—within which increases in the minimum wage raise employment and output despite increasing average wages.

In addition to this labor market mechanism, minimum wage policies affect firm behavior through the product market. When a higher minimum wage raises household income, product demand shifts outward. As shown above, prices are determined by households' willingness to pay for the differentiated good, which depends positively on income. An increase in income therefore raises the inverse demand curve faced by firms, increasing both equilibrium prices and marginal revenue.

The interaction of these two channels is central to the model.⁸By simultaneously reducing effective marginal labor costs and increasing marginal revenue through income-driven demand expansion, a binding minimum wage can induce firms to increase output and employment in the short run. When the minimum wage becomes sufficiently high, however, the cost effect dominates and firms reduce employment and output. The resulting non-monotonic relationship between minimum wages and employment motivates the numerical analysis in the next section.

⁸The non-monotonic relationship between minimum wages and employment emphasized here does not rely on knife-edge assumptions or specific functional forms. Rather, it arises generically when wage-setting distortions in the labor market interact with income-driven demand feedback in imperfectly competitive product markets.

4.5 Analytical Limitations and Transition to Numerical Analysis

The model developed in this section combines monopsonistic competition in the labor market with imperfect competition and income effects in the product market. Firms choose employment and output while accounting for an upward-sloping labor supply curve, endogenous household income, and a nonlinear inverse demand function derived from household optimization. While this framework allows us to characterize the key mechanisms through which minimum wages affect employment and output, it also leads to a system of equilibrium conditions that is highly nonlinear.

Equilibrium outcomes are jointly determined by firms' output choices, endogenous wages implied by household labor supply, and equilibrium prices determined by household demand. Importantly, product demand depends on household income, which is itself a function of wages and employment. This feedback loop implies that closed-form solutions for equilibrium output, employment, and prices are generally unavailable except under restrictive functional form assumptions.

These analytical limitations arise naturally from the interaction between labor market distortions, income-driven product demand, and firm optimization, rather than from any simplifying assumptions. Accordingly, the next section adopts a numerical approach to characterize equilibrium outcomes under alternative minimum wage levels and parameter configurations. This approach allows us to study how the effects of minimum wage policies vary systematically with production technology, labor supply elasticity, and the income sensitivity of product demand.

5 Quantitative Implications

This section quantifies the implications of the theoretical model by numerically solving firms' profit maximization problems under alternative minimum wage levels and structural parameter configurations. The analysis focuses on short-run equilibria, holding capital fixed and abstracting from long-run adjustments.

5.1 Parameterization and Baseline Configuration

It is useful to briefly summarize the roles of the key structural parameters in the model. The production side is characterized by the capital share δ and the CES curvature parameter ρ_p , which together determine the capital–labor substitution elasticity $\sigma = 1/(1 - \rho_p)$. Labor market behavior is governed by the labor supply elasticity parameter γ and the level of non-labor income Y_0 , which shape the degree of monopsonistic distortion. On the demand side, the parameter ρ_d controls the strength of income effects in product demand and thus the magnitude of the income-driven demand channel. The remaining parameters serve normalization or scale purposes.

Unless otherwise stated, parameters are fixed at the following baseline values: $A = 1$, $K = 1$, $r = 0.02$, $\delta = 0.5$, $\rho_p = 0$, $\rho_d = 0.6$, $\gamma = 1$, and $Y_0 = 0.2$. Alternative configurations vary one parameter at a time while holding all others fixed.

5.2 Numerical Method and Baseline Parameters

For each parameter configuration, we first compute the baseline equilibrium without a minimum wage. We then solve for equilibrium output, employment, wages, and prices over a grid of minimum wage levels. An *output-enhancing minimum wage interval* is defined as the set of minimum wages for which equilibrium output exceeds its baseline level.

Figure 1 illustrates the baseline equilibrium responses to changes in the minimum wage. When the minimum wage lies below the endogenous equilibrium wage, it is non-binding and has no effect on employment, output, or prices. This region corresponds to the standard case in which firms continue to set wages according to the upward-sloping labor supply curve.

As the minimum wage rises modestly above the equilibrium wage, it becomes binding and alters firms' marginal hiring incentives. In this range, employment and output both increase. This expansion reflects two complementary mechanisms. First, fixing the wage eliminates the wage-pushing term $Lw'(L)$ from firms' effective marginal labor costs, lowering the cost of expanding employment. Second, higher wages raise household income, which shifts product demand outward and increases firms' marginal revenue.

When the minimum wage increases further, however, the expansionary effects reverse. Although prices continue to rise due to higher labor costs and income-driven demand, the term $W_{\min} \frac{dL}{dQ}$ in the firm's first-order condition grows sufficiently large to dominate. As a result, marginal costs increase sharply, leading firms to contract output and employment. This non-monotonic response highlights the existence of an interior range of minimum wages that can raise employment and output in the short run, but also underscores that excessively high minimum wages are contractionary.

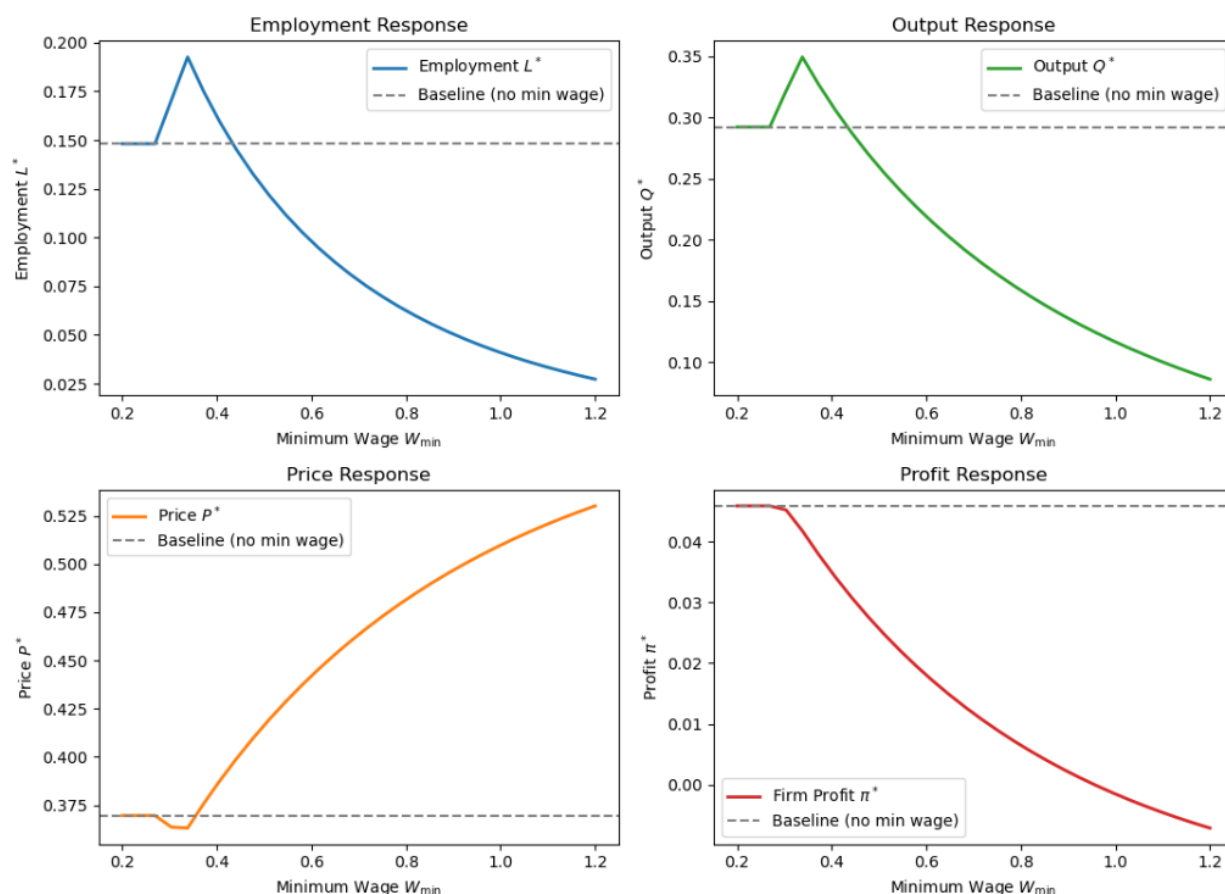


Chart 1: Baseline effects of minimum wage changes on employment, output, prices, and firm profits

Source: Author's numerical simulations.

5.3 Output-Enhancing Minimum Wage Intervals

Table 1 reports the ranges of minimum wages for which equilibrium output exceeds its baseline level across alternative production technologies, demand conditions, and labor supply elasticities.

Table 1: Output-Increasing Minimum Wage Intervals under Alternative Structural Parameters

$\sigma (\rho_p)$	δ	ρ_d	γ	Baseline Q	$W_{\min}^L$	$W_{\min}^H$	Interval Width
<i>Panel A. Capital–Labor Substitutability</i>							
0.56	0.50	0.60	1.00	0.317	0.309	0.528	0.219
0.71	0.50	0.60	1.00	0.340	0.301	0.469	0.168
1.00	0.50	0.60	1.00	0.366	0.276	0.385	0.109
1.43	0.50	0.60	1.00	0.387	0.259	0.318	0.059
2.50	0.50	0.60	1.00	0.410	0.225	0.250	0.025
<i>Panel B. Capital Intensity</i>							
1.00	0.20	0.60	1.00	0.241	0.309	0.494	0.185
1.00	0.35	0.60	1.00	0.298	0.292	0.452	0.160
1.00	0.50	0.60	1.00	0.366	0.276	0.385	0.109
1.00	0.65	0.60	1.00	0.451	0.259	0.318	0.059
1.00	0.80	0.60	1.00	0.569	0.234	0.259	0.025
<i>Panel C. Income Sensitivity of Demand</i>							
1.00	0.50	0.10	1.00	0.425	0.318	0.578	0.260
1.00	0.50	0.30	1.00	0.408	0.301	0.486	0.185
1.00	0.50	0.60	1.00	0.366	0.276	0.385	0.109
1.00	0.50	1.00	1.00	0.304	0.250	0.301	0.051
1.00	0.50	1.50	1.00	0.219	0.225	0.242	0.017
<i>Panel D. Labor Supply Elasticity</i>							
1.00	0.50	0.60	0.30	0.533	–	–	0.000
1.00	0.50	0.60	0.60	0.439	0.200	0.259	0.059
1.00	0.50	0.60	1.00	0.366	0.276	0.385	0.109
1.00	0.50	0.60	1.50	0.309	0.402	0.536	0.134
1.00	0.50	0.60	2.50	0.242	0.637	0.813	0.176

Notes: Each row reports equilibrium outcomes under a given parameter configuration. Baseline parameters are $\sigma = 1$, $\delta = 0.5$, $\rho_d = 0.6$, and $\gamma = 1$. The parameter varied within each panel is highlighted in **bold**. Interval width is defined as $W_{\min}^H - W_{\min}^L$. *Source:* Author’s numerical simulations.

6 Industry-Level Predictions and Exposure to Minimum Wage Policy

This section translates the quantitative implications of the theoretical model into testable industry-level predictions and motivates the classification scheme used in the empirical analysis. By map-

ping key structural parameters—labor market power, demand conditions, and production technology—to observable industry characteristics, the model generates clear predictions regarding heterogeneous employment responses to minimum wage increases across sectors.

6.1 Industry Characteristics and Employment Responses

Table 1 illustrates substantial heterogeneity in employment and output responses to minimum wage increases across alternative structural environments. These differences arise from the interaction of labor market power, product market demand, and production technology.

Capital–Labor Substitutability. Panel A shows that capital–labor substitutability plays a central role in shaping employment responses. When capital and labor are strong complements (low substitution elasticity σ), the range of output- and employment-enhancing minimum wages is wide. As substitutability increases, this interval contracts and eventually disappears, as firms respond to higher wages by substituting toward capital rather than expanding employment.

Capital Intensity. Panel B highlights the importance of capital intensity. In labor-intensive industries, labor costs constitute a large share of marginal costs, so alleviating the wage-pushing distortion generates a stronger expansion in hiring. In contrast, in capital-intensive industries, minimum wages affect a narrower margin of production decisions, limiting scope for employment growth.

Income Sensitivity of Demand. Panel C demonstrates that income effects in product demand critically shape employment responses. When demand is highly sensitive to household income (high ρ_d), minimum wage increases shift the inverse demand curve outward, reinforcing firms’ incentives to expand output and employment. This mechanism is particularly relevant for industries producing locally consumed, non-tradable goods.

Labor Supply Elasticity. Panel D shows that labor supply elasticity governs the degree of monopsony power. When labor supply is relatively inelastic (low γ), firms possess substantial wage-setting power, and minimum wages are more likely to relax the wage-pushing distortion. As labor supply becomes more elastic, monopsony power weakens and positive employment effects diminish.

Summary of Model-Based Predictions. Taken together, the model predicts that employment responses to minimum wage policies depend systematically on industry structure. Industries characterized by strong monopsony power, income-sensitive local demand, labor-intensive production, and limited capital–labor substitution are most likely to exhibit neutral or positive employment responses to moderate minimum wage increases. In contrast, capital-intensive, tradable, or high-skill industries are more likely to experience muted or negative employment effects, even at moderate policy levels.

6.2 Industry Classification and Wage-Based Exposure to Minimum Wage Policy

To connect these theoretical predictions to the data, industries are classified according to their exposure to minimum wage policy using observed wage distributions. We pool all province–industry–month observations over the sample period (33 months) and compute national average hourly wages for each industry, thereby abstracting from province-specific fluctuations and short-run shocks.

Industries with national average hourly wages at or below \$26 are classified as low-wage industries, those with wages between \$26 and \$35 as middle-wage industries, and those with wages above \$35 as high-wage industries. This relative classification is motivated by the bindingness of minimum wage regulations rather than by absolute poverty thresholds.

This wage-based classification provides a transparent empirical proxy for exposure to minimum wage policy. Low-wage industries are those in which statutory minimum wages are most likely to

bind directly, middle-wage industries exhibit partial or indirect exposure, and high-wage industries face limited direct exposure and therefore serve as a natural comparison group.

Finally, the remaining industry characteristics—labor monopsony, income sensitivity of demand, and labor intensity—are not imposed as regression variables but serve as organizing dimensions grounded in established theoretical and empirical literatures. The classification of labor monopsony follows the insight that limited worker mobility, weak outside options, and localized labor markets generate wage-setting power on the employer side (Manning, 2003, 2021). Income sensitivity of demand reflects whether an industry primarily produces locally consumed, non-tradable goods, for which wage increases translate into higher local purchasing power and outward shifts in product demand (Mian and Sufi (2014) and Autor et al. (2019)). Labor intensity captures differences in production technology and cost structure, consistent with standard labor demand and industrial organization frameworks (Hamermesh, 1993; Cabral, 2017).

These characteristics are therefore used as interpretive lenses rather than precise empirical treatments, allowing estimated employment effects to be understood through underlying economic mechanisms rather than purely statistical groupings.

Table 2: Industry Wage Classification Based on National Average Hourly Wages

Industry	National avg. hourly wage (\$)	Wage group
Accommodation and food services	19.16	Low-wage
Agriculture	24.60	Low-wage
Wholesale and retail trade	24.97	Low-wage
Business, building and other support services	25.20	Low-wage
Other services (except public administration)	28.28	Middle-wage
Information, culture and recreation	29.25	Middle-wage
Transportation and warehousing	31.60	Middle-wage
Services-producing sector	32.10	Middle-wage
Health care and social assistance	32.14	Middle-wage
Manufacturing	32.20	Middle-wage
Total, all industries	32.63	Middle-wage
Construction	33.49	Middle-wage
Goods-producing sector	34.72	Middle-wage
Finance, insurance, real estate, rental and leasing	37.47	High-wage
Educational services	38.27	High-wage
Public administration	42.05	High-wage
Professional, scientific and technical services	42.07	High-wage
Forestry, fishing, mining, quarrying, oil and gas	42.37	High-wage
Utilities	48.28	High-wage

Source: Author's calculations using Statistics Canada Table 14-10-0063-01.

Table 3: Model-Based Industry Predictions on Employment Responses to Minimum Wages

Industry	Low-wage (LW)	Labor monopsony (LM)	Income- sensitive demand (ID)	Labor- intensive (LI)	Predicted employment response
Accommodation and food services	✓	Δ	✓	✓	Neutral / Heterogeneous
Wholesale and retail trade	✓	✓	✓	✓	Positive / Neutral
Agriculture	✓	Δ	Δ	✓	Neutral / Heterogeneous
Business, building and support services	✓	✓	✓	✓	Positive / Neutral
Other services (except public administration)	Δ	✓	✓	✓	Small (±)
Information, culture and recreation	Δ	Δ	Δ	Δ	Small (±)
Transportation and warehousing	Δ	✓	✓	✓	Small (±)
Health care and social assistance	Δ	✓	✓	✓	Small (±)
Manufacturing	Δ	Δ	×	Δ	Small negative
Construction	Δ	Δ	×	✓	Small (±)
Finance, insurance, real estate, rental and leasing	×	Δ	✓	Δ	No systematic effect
Professional, scientific and technical services	×	Δ	Δ	✓	No systematic effect
Educational services	×	✓	✓	✓	No systematic effect
Forestry, fishing, mining, quarrying, oil and gas	×	Δ	×	×	No systematic effect
Utilities	×	×	×	×	No systematic effect
Public administration	×	×	×	×	No systematic effect

Notes: LW captures whether minimum wages are likely to bind and primarily governs the expected magnitude of employment effects. LM, ID, and LI indicate labor market power, income sensitivity of demand, and labor intensity, respectively, and shape the potential direction of effects. A checkmark denotes substantial relevance of the mechanism, Δ partial or context-dependent relevance, and limited relevance. Industries marked with * are characterized by institutionally determined wages rather than market-based wage setting. Predicted employment responses are stated for a high minimum wage environment, as in the Canadian context. *Source:* Author's numerical simulations and author's interpretation.

7 Framework for Estimating the Effects of Recent Minimum Wage Changes

This section describes the empirical framework used to estimate the employment effects of recent minimum wage increases. The analysis adopts a standard program evaluation approach, exploiting variation in the timing and magnitude of province-level minimum wage changes and heterogeneity in industry-level exposure to these policies. The primary objective is to identify dynamic and mid-run employment responses across industries following minimum wage shocks.

7.1 Baseline Difference-in-Differences Specification

We begin with a baseline difference-in-differences (DID) framework that relates employment outcomes to provincial minimum wage increases occurring in 2023 or later. The baseline specification is estimated separately for each of the 19 industries in the sample, yielding a matrix of industry-specific DID estimates. This design is intended to provide a reduced-form benchmark for heterogeneity across industries, rather than a fully dynamic causal estimate.

Treatment is defined based on the first occurrence of a statutory minimum wage increase in 2023 or later. A province is classified as treated from the first such increase onward, and all subsequent months are considered post-treatment (once-treated-always-treated). For each industry i , we estimate the following two-way fixed effects specification:

$$Y_{i,s,t} = \beta_i (\text{Treated}_s \times \text{Post}_{s,t}) + \alpha_s + \alpha_t + \varepsilon_{i,s,t}, \quad (18)$$

where $Y_{i,s,t}$ denotes the employment outcome in industry i , province s , and month t . The indicator $\text{Post}_{s,t}$ equals one for all months following the first post-2023 minimum wage increase in province s , and zero otherwise. Province fixed effects α_s control for time-invariant differences across provinces, while time fixed effects α_t absorb common macroeconomic shocks. Standard errors are clustered at the province level.

This baseline DID specification captures, for each industry, the average difference in employment outcomes before and after the first post-2023 minimum wage increase, relative to provinces that have not yet experienced such an increase. However, several features of the Canadian policy environment limit the credibility of this design as a causal estimator.

First, due to the highly coordinated nature of minimum wage policy in Canada, as it shows in the picture, most provinces implement increases within a relatively short time window.

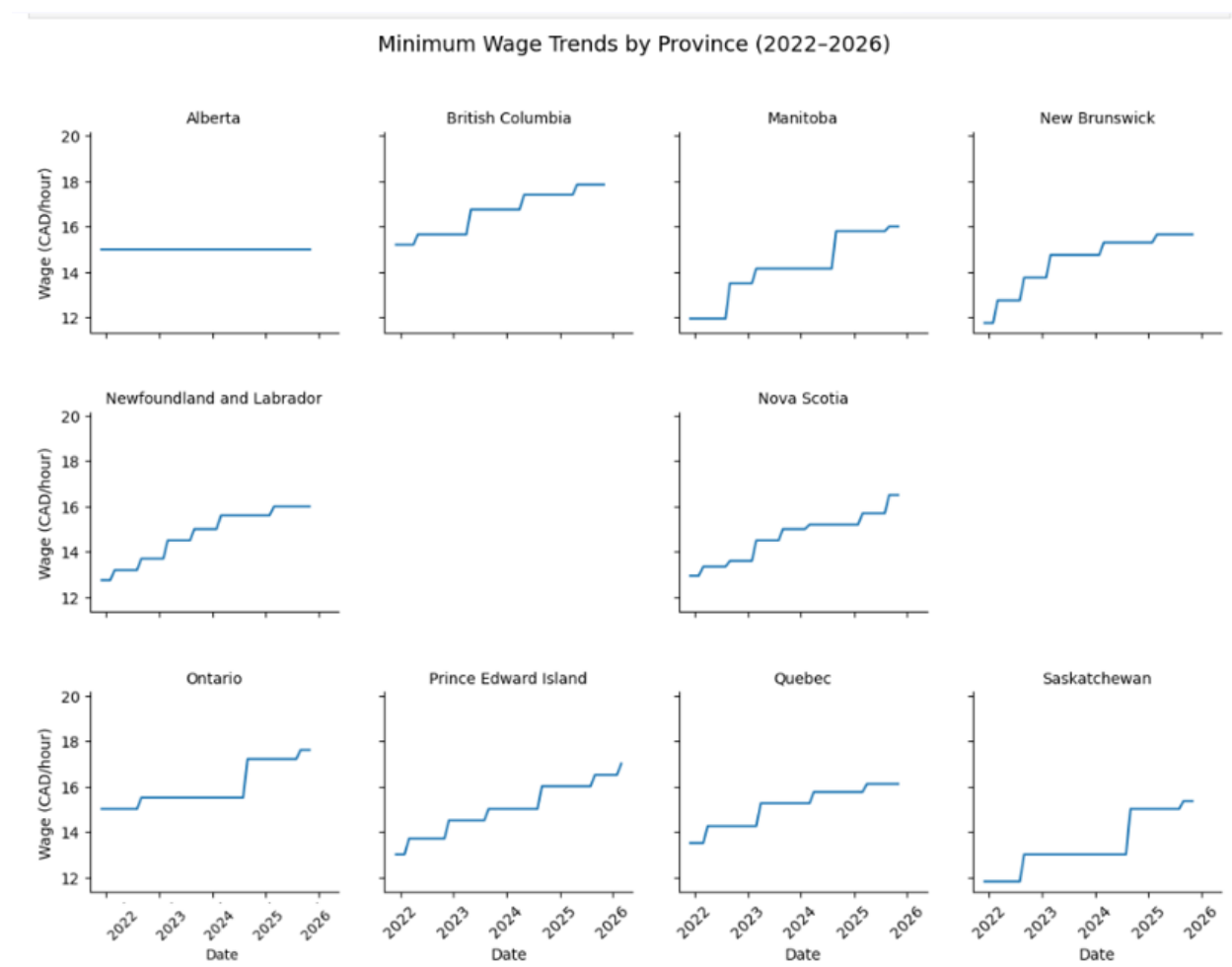


Chart 2: Frequent minimum wage adjustments in Canada

Source: Author's calculations using Employment and Social Development Canada (ESDC) data.

As a result, the set of untreated provinces shrinks rapidly over the sample period. In practice, Alberta emerges as the primary remaining control province during much of the post-2023 period. Consequently, the estimated DID coefficients increasingly reflect differential employment trends

between treated provinces and Alberta, rather than comparisons across multiple untreated jurisdictions. To the extent that Alberta exhibits distinct labor market dynamics—such as a stronger resource cycle, different sectoral composition, or province-specific trends—the baseline DID estimates may conflate minimum wage effects with Alberta-specific economic trajectories.

Second, minimum wage adjustments were already frequent in 2022, raising concerns about dynamic policy spillovers. If minimum wage increases implemented in 2022 continue to affect employment outcomes in 2023 and beyond, the pre-treatment period used in the baseline DID specification is no longer clean. In this case, the estimated treatment effects may be contaminated by lingering impacts of earlier policy changes, violating the standard parallel trends assumption.

Taken together, these considerations imply that the baseline DID results should be interpreted as descriptive, industry-specific benchmarks rather than definitive causal estimates. To address these limitations, we extend the analysis to a stacked event-study and triple-difference framework that explicitly models dynamic adjustment paths and exploits within-province, within-industry variation in policy exposure.

Table 4: Baseline Difference-in-Differences Estimates by Industry

Industry	DID Coefficient	Std. Error	<i>p</i> -value	<i>N</i>
Professional, scientific and technical services	-0.0237	0.0135	0.079*	330
Transportation and warehousing	-0.0172	0.0113	0.129	330
Information, culture and recreation	-0.0125	0.0085	0.141	330
Manufacturing	-0.0096	0.0069	0.164	330
Accommodation and food services	-0.0082	0.0068	0.228	330
Construction	0.0084	0.0078	0.280	330
Wholesale and retail trade	0.0070	0.0058	0.225	330
Health care and social assistance	0.0043	0.0034	0.203	330
Public administration	-0.0020	0.0031	0.522	330
Goods-producing sector	-0.0040	0.0077	0.606	330
Educational services	-0.0034	0.0080	0.675	330
Utilities	-0.0037	0.0089	0.680	326
Finance, insurance, real estate	-0.0026	0.0083	0.748	330
Services-producing sector	-0.0008	0.0026	0.752	330
Agriculture	-0.0122	0.0412	0.767	330
Total, all industries	-0.0022	0.0024	0.363	330

Notes: Each coefficient reports the estimate from a simple difference-in-differences regression of industry-level employment rates on a treated $\times$ post indicator, where treated provinces are those experiencing at least one minimum wage increase after January 2023. All specifications include province and month fixed effects. Standard errors are clustered at the province level. *Source:* Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

7.2 Industry-Specific Stacked Difference-in-Differences with Short-Run Dynamics and Medium-Run Exposure

To capture dynamic employment responses to minimum wage increases in the presence of staggered provincial policy changes, we implement an industry-specific stacked event-study difference-in-differences (DID) design. Rather than relying on a single pooled specification, the model is estimated separately for each industry, allowing employment responses to vary flexibly across sectors.

For each province-level minimum wage increase occurring after 2023, we construct a symmetric event window around the policy implementation date and stack these windows across events. This approach follows the stacked DID framework and ensures that identification exploits variation in the timing and magnitude of minimum wage changes while avoiding contamination from overlapping treatment episodes.

The estimating equation for a given industry i takes the following form:

$$Y_{i,s,t} = \sum_{k \neq k_0} \beta_k (\mathbf{1}\{t - t_0 = k\} \times \text{Dose}_s) + \delta \Delta \log W_{s,t} + \alpha_s + \alpha_t + \varepsilon_{i,s,t}, \quad (19)$$

where $Y_{i,s,t}$ denotes the employment rate in industry i , province s , and month t . The indicator $\mathbf{1}\{t - t_0 = k\}$ captures event time k relative to the minimum wage increase occurring at t_0 , with k_0 denoting the omitted baseline period. The variable Dose_s measures the proportional size of the minimum wage increase associated with the event, allowing treatment intensity to vary continuously across provinces. The term $\Delta \log W_{s,t}$ captures cumulative minimum wage growth relative to a province-specific pre-policy baseline and is included to control for medium-run exposure to minimum wage changes beyond the short-run event window.

Identification in this framework exploits variation in the timing and magnitude of minimum wage increases across provinces. For each event, treated provinces within a given event window are compared to a set of “clean control” provinces that do not experience any minimum wage change during the same period. Province fixed effects α_s control for time-invariant provincial characteristics, while time fixed effects α_t absorb aggregate shocks common to all provinces and

industries.

To construct valid control groups, observations belonging to other treatment windows are excluded from the control sample in each stacked dataset. This “clean control” or “hole-punching” procedure ensures that control observations are not affected by minimum wage changes in nearby periods.

The model is estimated separately for each industry, producing industry-specific event-study estimates under a common identification strategy.

7.3 Short-Run and Medium-Run Parameters

A key feature of the empirical specification is the joint inclusion of event-time parameters $\{\beta_k\}$ and a cumulative minimum wage measure $\Delta \log W_{s,t}$ with associated coefficient δ . This structure is designed to separate localized adjustment dynamics surrounding discrete minimum wage changes from more persistent, medium-run employment responses associated with sustained increases in the wage floor during periods of repeated policy intervention.

The event-time parameters $\{\beta_k\}$ capture short-run dynamics around each minimum wage increase. These coefficients trace employment responses within a narrow event window surrounding the policy implementation date, allowing for anticipatory effects, delayed adjustment, and post-treatment dynamics. Identification of $\{\beta_k\}$ relies on comparisons between treated provinces and “clean control” provinces within the same event window, thereby isolating localized, short-horizon responses to a given policy shock.

In contrast, the cumulative minimum wage term $\Delta \log W_{s,t}$, with coefficient δ , captures the persistent component of minimum wage exposure associated with sustained increases in the wage floor over time. Constructed as the change in the logarithm of the minimum wage relative to a province-specific pre-policy baseline, $\Delta \log W_{s,t}$ measures the extent to which statutory minimum wages have been permanently and monotonically raised during a period of repeated policy interventions. The coefficient δ therefore reflects a medium-run employment response to sustained wage increases, rather than a long-run steady-state effect.

Including the cumulative exposure term $\Delta \log W_{s,t}$ plays an important identification role. Because minimum wage adjustments in Canada are staggered and frequently repeated, earlier policy changes may generate persistent shifts in employment levels that extend beyond the narrow event windows surrounding subsequent increases. The definition of the event window is therefore an important component of the identification strategy.

The event window is defined to span two months prior to and three months following each minimum wage increase ($k \in [-2, 3]$). This choice reflects institutional features of minimum wage setting in Canada and serves several practical purposes. First, given the frequency of minimum wage adjustments over the sample period, this window length avoids overlap between adjacent event windows, ensuring that each policy change can be analyzed in isolation. Second, the inclusion of pre-treatment periods allows for an assessment of pre-trends and anticipatory responses, while the post-treatment horizon is sufficiently long to capture immediate adjustment dynamics and announcement effects surrounding policy implementation. Third, restricting the event window to a relatively narrow range preserves a substantial set of observations outside treatment windows, ensuring that a large and stable pool of “clean control” observations remains available for identification.

By jointly estimating $\Delta \log W_{s,t}$ with the event-time parameters $\{\beta_k\}$, the specification absorbs these persistent components of policy exposure and prevents them from being misattributed to the short-run event-time coefficients. Absent this control, the estimated β_k may incorrectly capture accumulated effects of prior minimum wage increases rather than true dynamic responses to the current policy shock.

Taken together, this joint specification yields a clear decomposition of employment responses: the event-time coefficients $\{\beta_k\}$ capture localized short-run adjustment dynamics around discrete policy changes, while the cumulative coefficient δ summarizes medium-run employment responses to sustained increases in the statutory wage floor.

7.4 Inference and Clustering at the Provincial Level

Inference in the stacked difference-in-differences framework requires accounting for within-province dependence in the regression residuals. Minimum wage policies in Canada are legislated and implemented at the provincial level, and all industries within a province are exposed to the same policy change at the same time. As a result, unobserved shocks affecting employment outcomes are likely to be correlated across industries and over time within provinces.

To address this concern, all standard errors are clustered at the province level. This clustering choice aligns the inference with the level of treatment assignment and accounts for both serial correlation in employment outcomes over time and cross-industry dependence induced by common provincial policy shocks. Clustering at lower levels, such as the industry or province–industry level, would fail to capture this correlation structure and may lead to overstated statistical significance.

A potential limitation of province-level clustering in the Canadian context is the small number of clusters. Because Canada has only ten provinces, conventional cluster-robust variance estimators may suffer from finite-sample bias, potentially leading to over-rejection of the null hypothesis, particularly in joint tests of dynamic treatment effects in event-study specifications.

To address this concern, Appendix A reports inference based on wild cluster bootstrap procedures designed for settings with a small number of clusters. These bootstrap-based p -values provide a robustness check on the conventional cluster-robust inference. Across industries, the bootstrap results yield qualitatively similar conclusions for both short-run dynamic responses and medium-run exposure effects, supporting the validity of the main findings.

7.5 Result: Short-Run Adjustment Dynamics

Short-run employment responses are evaluated using a joint Wald test of the event-time coefficients within each industry. This test assesses whether employment exhibits statistically detectable dynamic adjustment around minimum wage increases, even when individual event-time coefficients are imprecisely estimated. Importantly, the joint test is designed to identify the existence of

short-run dynamics rather than to characterize their direction or magnitude.

The results indicate strong evidence of short-run adjustment in several industries. In accommodation and food services, agriculture, educational services, health care and social assistance, manufacturing, professional, scientific and technical services, and public administration, the null hypothesis that all event-time coefficients are jointly equal to zero is rejected at conventional significance levels. These findings imply that employment responds dynamically around minimum wage increases in these sectors.

By contrast, a number of industries—including construction, finance and real estate, wholesale and retail trade, business support services, and other services—exhibit neither statistically significant joint tests nor systematically significant event-time coefficients. These sectors therefore serve as an implicit falsification check, suggesting that the estimated short-run dynamics are not driven by common macroeconomic shocks or spurious correlation with policy timing.

While the joint test establishes the presence of dynamic adjustment, the direction and timing of short-run effects are conveyed by the individual event-time coefficients. These coefficients are estimated relative to the baseline period $k = -1$, corresponding to the month immediately preceding the policy change. Across industries, the event-study paths reveal heterogeneous adjustment patterns. For example, agriculture displays a contemporaneous decline in employment at the time of the policy change, while educational services shows a positive employment response two months after implementation. In professional, scientific and technical services, employment declines persist for several months following treatment, consistent with gradual adjustment rather than immediate displacement.

Event-time coefficients in pre-treatment periods provide additional validation of the empirical design. In particular, coefficients at $k = -2$ are generally small and statistically insignificant across industries, indicating an absence of differential pre-trends prior to the policy change. This pattern supports the parallel trends assumption underlying the difference-in-differences framework and strengthens the causal interpretation of the estimated post-treatment dynamics.

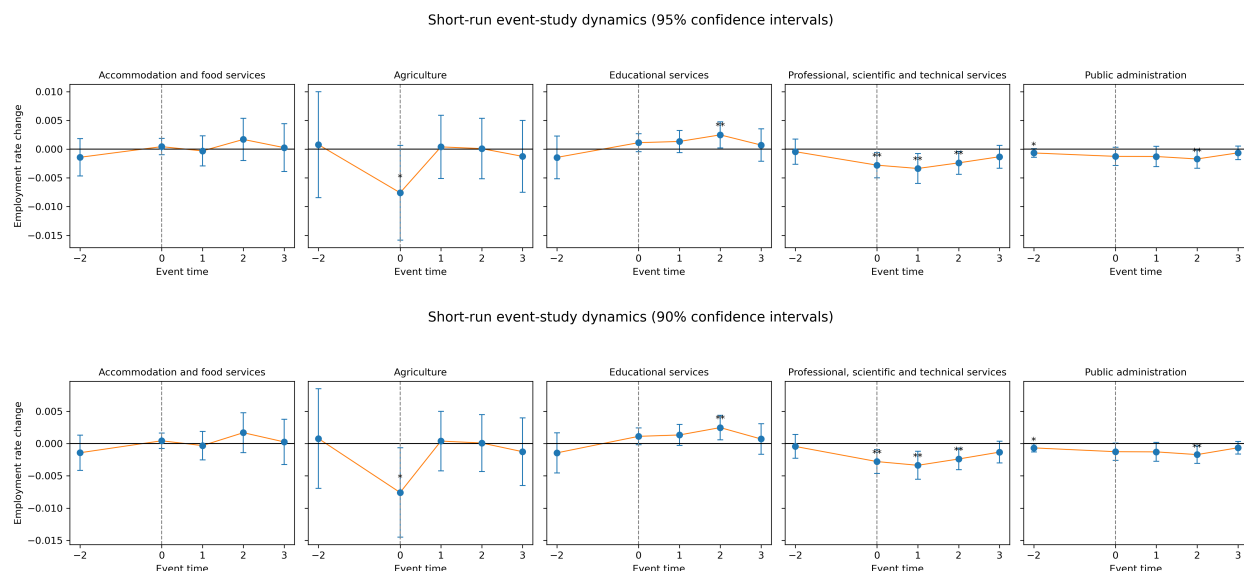


Chart 3: Short-Run Event-Study Dynamics of Employment Around Minimum Wage Increases

Notes: This figure plots estimated event-time coefficients from industry-level event-study regressions examining short-run employment responses to minimum wage increases. Panel A reports point estimates with 95% confidence intervals, while Panel B reports point estimates with 90% confidence intervals. Event time is measured in months relative to the policy implementation date, with $k = -1$ (the month immediately preceding the minimum wage increase) serving as the omitted reference category. Vertical dashed lines indicate the timing of the policy change. Confidence intervals are constructed using heteroskedasticity-robust standard errors clustered at the province–industry level. Asterisks denote statistical significance at conventional levels. Pre-treatment coefficients provide a visual diagnostic of differential pre-trends, while post-treatment coefficients illustrate heterogeneous short-run adjustment dynamics across industries. *Source:* Authors’ calculations based on Statistics Canada data.

Table 5: Short-Run Dynamic Employment Responses Around Minimum Wage Increases

Industry	$k = -2$	$k = 0$	$k = 1$	$k = 2$	$k = 3$	Joint Test
Accommodation and food services	-0.00142	0.00043	-0.00031	0.00169	0.00026	Yes***
Agriculture	0.00078	-0.00759*	0.00039	0.00009	-0.00126	Yes***
Educational services	-0.00144	0.00112	0.00133	0.00248**	0.00071	Yes**
Professional, scientific and technical services	-0.00045	-0.00279**	-0.00337**	-0.00239**	-0.00132	Yes***
Public administration	-0.00067*	-0.00126	-0.00129	-0.00171**	-0.00065	Yes***

Notes: Each column reports event-time coefficients from stacked difference-in-differences regressions, with $k = -1$ omitted as the baseline period. Coefficients represent short-run employment rate changes. Standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Author’s calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

7.6 Result: Medium-Run Cumulative Effects

In addition to short-run dynamics, the specification identifies medium-run employment responses associated with sustained increases in the minimum wage.⁹ These effects are captured by the coefficient on cumulative minimum wage growth relative to a province-specific pre-policy baseline. Unlike the event-time coefficients, this term reflects persistent, non-window exposure to a permanently higher wage floor during a period of repeated policy intervention.

The estimated cumulative coefficient is negative and statistically significant in accommodation and food services, professional, scientific and technical services, and transportation and warehousing. These results suggest that, conditional on short-run dynamics, continued upward pressure on the wage floor is associated with lower employment levels in these sectors. The magnitude of these estimates indicates economically meaningful medium-run adjustments in industries that are relatively labor intensive or more exposed to low-wage employment.

Importantly, medium-run effects are not uniform across industries. In many sectors—including construction, finance, wholesale and retail trade, and the aggregate economy—the cumulative coefficient is small and statistically indistinguishable from zero. This pattern reinforces the view that employment responses to minimum wage increases are highly context dependent and concentrated in a subset of industries.

⁹Throughout the paper, “medium-run” refers to cumulative employment responses observed within the sample window characterized by repeated minimum wage adjustments. These effects should not be interpreted as long-run steady-state outcomes, as capital adjustment, firm entry and exit, and worker reallocation typically operate over longer horizons than those observed in the data. Moreover, the cumulative exposure term is measured as the log change in the minimum wage relative to a pre-policy baseline, and its coefficient therefore differs in both scale and interpretation from the short-run event-time coefficients, which are scaled by event-specific percentage changes. As a result, the two sets of coefficients should not be compared directly.

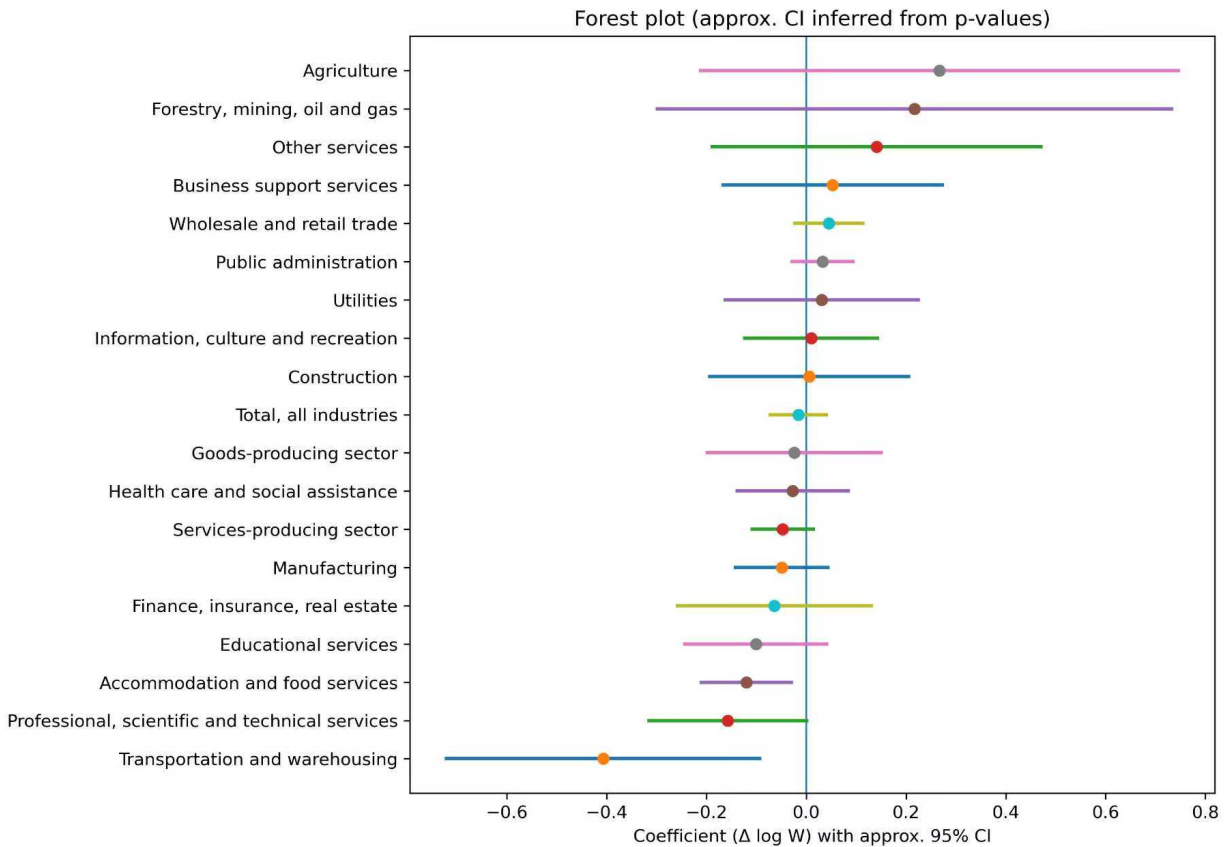


Chart 4: Industry-Level Medium-Run Employment Effects of Minimum Wage Increases

Source: Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

Table 6: Medium-Run Employment Effects of Sustained Minimum Wage Increases

Industry	Coefficient ($\Delta \log W$)	<i>p</i> -value
Accommodation and food services	-0.120***	0.009
Agriculture	0.267	0.274
Business support services	0.053	0.636
Construction	0.006	0.953
Educational services	-0.101	0.163
Finance, insurance, real estate	-0.064	0.518
Forestry, mining, oil and gas	0.217	0.409
Goods-producing sector	-0.024	0.787
Health care and social assistance	-0.027	0.634
Information, culture and recreation	0.010	0.883
Manufacturing	-0.049	0.304
Other services	0.141	0.401
Professional, scientific and technical services	-0.157*	0.051
Public administration	0.033	0.287
Services-producing sector	-0.047	0.132
Transportation and warehousing	-0.407***	0.011
Utilities	0.031	0.753
Wholesale and retail trade	0.045	0.196
Total, all industries	-0.016	0.576

Notes: Each row reports the coefficient on cumulative minimum wage growth relative to the pre-2023 baseline. This term captures medium-run employment responses to sustained increases in the wage floor outside event windows.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *Source:* Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

7.7 Interpretation and Identification

Unlike traditional triple-difference designs that rely on comparisons between explicitly defined target and non-target worker groups, the present analysis does not impose *ex ante* assumptions regarding which industries or workers are affected by minimum wage changes. Instead, heterogeneity in employment responses is identified directly from the data through industry-level variation.

This approach allows the estimates to capture both direct effects in low-wage, labor-intensive industries and indirect adjustments in other sectors, such as changes in labor demand driven by cost pass-through, substitution, or output responses. A credible specification in this setting should therefore display economically meaningful responses primarily in industries plausibly exposed to minimum wage regulation, while exhibiting limited effects in sectors with minimal exposure.

The empirical patterns documented in this section are broadly consistent with this expectation: statistically significant short-run dynamics and medium-run cumulative effects are concentrated in a subset of industries with higher exposure to the minimum wage, while many other sectors exhibit effects that are small and statistically indistinguishable from zero.

The next section evaluates the robustness of these findings by subjecting the main results to a series of diagnostic and sensitivity analyses designed to address alternative interpretations related to policy timing, macroeconomic co-movement, and cross-province spillovers.

8 Robustness Checks and Extensions

This section examines the robustness of the main empirical findings to alternative specifications and identification concerns. The analyses are designed to assess whether the estimated employment responses are sensitive to the spacing of minimum wage events, policy timing, spatial correlation across provinces, or sampling variation.

Importantly, these exercises are not intended to uncover new effects, but rather to evaluate the stability and interpretability of the results presented in the previous section.

8.1 Robustness to Macro Controls

Robustness to macroeconomic controls is assessed by augmenting the baseline matrix difference-in-differences specification, estimated separately for each of the 19 industries, with a set of province-level variables capturing contemporaneous macroeconomic conditions. Specifically, we include monthly retail sales, which proxy both local demand conditions and overall economic activity (GDP), the consumer price index (CPI) to account for inflationary pressures, and provincial population to capture changes in labor market scale and demographic composition.

Overall, the results remain broadly consistent with those reported in the baseline specification. Across industries, the estimated treatment effects exhibit similar magnitudes and dynamic patterns after the inclusion of macro controls. While statistical significance is attenuated for some coefficients, a small number of industries continue to display effects at the 5 percent or 10 percent levels, and in all such cases the estimated signs are aligned with those obtained in the baseline regressions.

Importantly, the industries that drive the main baseline findings continue to feature prominently in the macro-controlled specification. Sectors such as accommodation and food services, professional, scientific and technical services, and transportation and warehousing retain statistically meaningful responses, particularly in the medium run, even after conditioning on macroeconomic variables. This pattern suggests that the baseline results are not solely driven by coincident movements in aggregate demand or inflation.

At the same time, we do not adopt the macro-controlled specification as the preferred baseline. The included macroeconomic variables may constitute potential bad controls, as minimum wage policies can directly or indirectly affect local economic activity, prices, and population dynamics. Conditioning on such outcomes may therefore absorb part of the policy-induced adjustment process and mechanically attenuate estimated employment effects. The robustness results instead indicate that province fixed effects and common time fixed effects already capture the dominant sources of aggregate macroeconomic variation at the monthly frequency.

Taken together, this robustness exercise indicates that the main conclusions are not driven by omitted macroeconomic factors, while reinforcing the interpretation of the baseline matrix DID

estimates as the most transparent representation of policy-induced employment adjustments.

Table 7: Medium-Run Employment Effects with Macro Controls

Industry	Coefficient ($\Delta \log W$)	p -value
Accommodation and food services	-0.173***	0.000
Manufacturing	-0.103**	0.039
Professional, scientific and technical services	-0.133*	0.071
Services-producing sector	-0.079**	0.013
Transportation and warehousing	-0.451***	0.004
Total, all industries	-0.064*	0.059

Notes: Each row reports the coefficient on cumulative minimum wage growth relative to the pre-2023 baseline from the matrix DID specification augmented with macro controls. Controls include provincial retail sales, CPI, and population. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *Source:* Author’s calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

8.2 Pre-trend Diagnostics and Alternative Baselines

A key identifying assumption of the stacked event-study design is that, absent minimum wage increases, treated provinces would have followed parallel employment trends relative to the clean-control provinces within the event window. To assess this assumption, we implement two complementary pre-trend diagnostics.

First, under the baseline event-study normalization (with $k = -1$ omitted), we test whether the immediate pre-period coefficient at $k = -2$ is statistically different from zero within each industry. The results provide limited evidence of pre-trends: only *public administration* exhibits a statistically significant deviation ($p = 0.0137$), while *health care and social assistance* is marginally significant ($p = 0.0620$). The remaining industries show no meaningful pre-treatment departures.

Second, because inference on a single lead period can be sensitive to baseline normalization and potential anticipation effects, we re-normalize the event-study by setting the implementation month ($k = 0$) as the omitted period and conduct a joint Wald test of the two pre-period coefficients at $k = -1$ and $k = -2$. Under this stricter diagnostic, *health care and social assistance* and the *services-producing sector* reject the joint null at conventional levels ($p = 0.0365$ and $p = 0.0461$, respectively), while *professional, scientific and technical services* shows marginal evidence of pre-

deviations ($p = 0.0986$). Importantly, the industries that reject the joint pre-trend test are generally not the same sectors driving the main post-treatment findings, suggesting that the baseline results are not systematically explained by differential pre-trends.

That said, we treat *professional, scientific and technical services* with additional caution, as the marginally significant joint pre-test may indicate some persistence of pre-period deviations into the post-period. At the same time, such patterns are not unexpected when conducting multiple joint tests across 19 industries, and they may also reflect anticipatory adjustments: in Canada, minimum wage changes are typically announced several months before implementation, so some degree of pre-implementation response is plausible and consistent with an announcement-effect interpretation rather than a violation of the identifying assumption.¹⁰

Table 8: Short-Run Pre-trend Test

Industry	Wald statistic	p -value
Public administration	6.072	0.014
Health care and social assistance	3.483	0.062
Goods-producing sector	2.394	0.122
Transportation and warehousing	1.928	0.165
Total, all industries	1.279	0.258
Services-producing sector	1.078	0.299
Construction	0.472	0.492
Other services (except public administration)	0.417	0.518
Educational services	0.343	0.558
Utilities	0.327	0.568
Accommodation and food services	0.261	0.609
Business, building and other support services	0.235	0.628
Finance, insurance, real estate, rental and leasing	0.210	0.647
Wholesale and retail trade	0.155	0.694
Professional, scientific and technical services	0.092	0.761
Manufacturing	0.087	0.768
Forestry, fishing, mining, quarrying, oil and gas	0.085	0.771
Information, culture and recreation	0.008	0.930
Agriculture	0.007	0.936

Notes: The table reports Wald tests of the null hypothesis that the event-study coefficient in the pre-treatment period is zero. The pre-treatment period ($k = -2$) is fixed by design and therefore not reported separately. *Source:* Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

¹⁰In Canada, provincial minimum wage increases are typically announced several months prior to their effective dates. This institutional feature allows for potential anticipation effects in employment outcomes before implementation.

Table 9: Joint Pre-trend Test

Industry	Wald statistic	<i>p</i> -value
Health care and social assistance	6.622	0.037
Services-producing sector	6.153	0.046
Professional, scientific and technical services	4.633	0.099
Public administration	3.903	0.142
Goods-producing sector	3.301	0.192
Other services (except public administration)	2.308	0.315
Total, all industries	2.173	0.337
Accommodation and food services	2.096	0.351
Manufacturing	2.007	0.367
Agriculture	1.451	0.484
Utilities	1.059	0.589
Construction	1.003	0.606
Information, culture and recreation	0.934	0.627
Wholesale and retail trade	0.834	0.659
Educational services	0.676	0.713
Transportation and warehousing	0.509	0.775
Forestry, fishing, mining, quarrying, oil and gas	0.313	0.855
Business, building and other support services	0.305	0.859
Finance, insurance, real estate, rental and leasing	0.204	0.903

Notes: This table reports joint Wald tests of the null hypothesis that all pre-treatment event-study coefficients are jointly zero. The pre-treatment periods ($k = -1, -2$) are fixed by design and therefore omitted from the table. *Source:* Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

8.3 Robustness: Medium-Run Effects by Inter-Event Spacing

As a robustness check on the linear specification of medium-run cumulative effects, we allow the long-run response to vary by the length of time between consecutive minimum wage increases. Specifically, minimum wage changes are divided into two groups based on whether the time until the next policy change in the same province is shorter or longer than the median inter-event gap of 12 months.

Table 10 reports cumulative employment effects separately for short-gap and long-gap events, along with an F -test of the null hypothesis that the two coefficients are equal. Across the majority of industries, the null hypothesis cannot be rejected, indicating no statistically meaningful difference in medium-run employment responses across gap lengths.

These results suggest that the estimated cumulative effects are not driven by the spacing of minimum wage changes and provide support for the linear accumulation assumption embedded in the baseline specification. In other words, employment responses appear to depend primarily on the cumulative magnitude of minimum wage growth rather than on whether increases occur in rapid succession or are more widely spaced.

A small number of sectors—such as other services and the aggregate economy—exhibit marginal evidence of differential effects across gap lengths. However, these cases are limited and do not display a consistent pattern across industries. Overall, the findings indicate that allowing for heterogeneity by inter-event spacing does not materially alter the baseline conclusions regarding medium-run employment effects.

Table 10: Medium-Run Employment Effects by Inter-Event Gap Length

Industry	Short Gap ($\Delta \log W$)	Long Gap ($\Delta \log W$)	H_0 : Equal Effects (p -value)
Accommodation and food services	-0.120***	-0.140	0.869
Agriculture	0.265	0.318	0.840
Business support services	0.048	0.185	0.415
Construction	0.008	-0.072	0.613
Educational services	-0.104	-0.014	0.100
Finance, insurance, real estate	-0.068	0.053	0.540
Forestry, mining, oil and gas	0.206	0.537*	0.173
Goods-producing sector	-0.025	-0.021	0.965
Health care and social assistance	-0.031	0.079	0.126
Information, culture and recreation	0.022	-0.330	0.097
Manufacturing	-0.047	-0.084	0.673
Other services	0.152	-0.176	0.045
Professional, scientific and technical services	-0.161**	-0.028	0.208
Public administration	0.033	0.042	0.775
Services-producing sector	-0.049	-0.000	0.110
Total, all industries	-0.017	0.019	0.060
Transportation and warehousing	-0.413***	-0.235*	0.196
Utilities	0.032	0.003	0.669
Wholesale and retail trade	0.044	0.095*	0.367

Notes: “Short gap” and “Long gap” correspond to cumulative minimum wage effects estimated separately for events followed by a shorter-than-median or longer-than-median interval before the next minimum wage increase. The median inter-event gap is 12 months. The final column reports p -values from an F -test of the null hypothesis that the two cumulative effects are equal. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *Source*: Author’s calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

8.4 Robustness: Placebo Timing Test

As a second robustness check, we conduct a placebo timing test to assess whether the estimated short-run dynamics are mechanically generated by the stacked event-study design or by spurious correlation with policy timing. Specifically, we reassign each minimum wage increase to a fictitious implementation date four months prior to the actual policy change and re-estimate the stacked difference-in-differences specification using these placebo events.

Table 11 compares joint Wald tests of the event-time coefficients from the baseline specification and the placebo timing test. In the baseline results, several industries exhibit strong evidence of short-run dynamic adjustment around minimum wage increases. For a number of these industries—including manufacturing, public administration, educational services, and utilities—the corresponding placebo joint tests fail to reject the null hypothesis of no dynamic response.

At the same time, the placebo exercise does not uniformly eliminate all statistically significant joint tests. A small number of industries that are insignificant in the baseline specification display significant placebo responses, while in a few highly exposed sectors—such as accommodation and food services and agriculture—the joint test remains statistically significant under the placebo timing, albeit with a noticeable reduction in test statistics.

This pattern suggests that the short-run dynamics identified in the main specification are not mechanically induced by the empirical design, but are also subject to short-run volatility in certain industries. Importantly, industries exhibiting significant placebo responses do not display a systematic correspondence with those identified in the baseline analysis, and the direction and magnitude of placebo event-time coefficients do not resemble the baseline dynamic paths.

Overall, the placebo timing test supports the interpretation that the short-run adjustment patterns documented in the main analysis are closely linked to the true timing of minimum wage changes, while also indicating that short-run estimates—particularly in highly exposed or volatile sectors—should not be over-interpreted in isolation. The results therefore strengthen the credibility of the identification strategy without implying that placebo effects must be uniformly absent across all industries.

Table 11: Placebo Timing Test: Joint Wald Tests for Short-Run Dynamics

Industry	Main F	Main p	Placebo F	Placebo p
Accommodation and food services	48.70	0.000	15.71	0.008
Manufacturing	22.16	0.001	3.91	0.563
Agriculture	20.93	0.001	19.95	0.001
Public administration	18.70	0.002	5.47	0.362
Health care and social assistance	18.63	0.002	13.83	0.017
Professional, scientific and technical services	17.32	0.004	12.21	0.032
Educational services	12.47	0.029	5.07	0.408
Goods-producing sector	9.88	0.079	21.68	0.001
Utilities	8.26	0.143	1.69	0.890
Services-producing sector	8.08	0.152	9.47	0.092
Information, culture and recreation	7.75	0.170	1.95	0.856
Wholesale and retail trade	7.46	0.189	4.18	0.524
Finance, insurance, real estate	6.74	0.241	4.14	0.529
Transportation and warehousing	6.37	0.272	2.34	0.801
Total, all industries	5.83	0.324	5.16	0.397
Other services	4.58	0.470	25.52	0.000
Forestry, mining, oil and gas	3.91	0.562	4.69	0.455
Business support services	3.70	0.594	8.47	0.132
Construction	1.81	0.875	21.79	0.001

Notes: The table compares joint Wald tests of event-time coefficients from the baseline specification and a placebo timing test in which minimum wage events are shifted four months earlier. Rejection of the joint null indicates the presence of dynamic employment responses around the assigned event date. *Source:* Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

8.5 Robustness: Permutation Inference for the Medium-Run Cumulative Effect

To assess the robustness of the estimated medium-run cumulative minimum wage effect, we implement permutation inference for the long-run coefficient on $\Delta \log W$. Specifically, we permute $\Delta \log W$ across provinces *within each month*, re-estimate the industry-specific stacked specifications, and compute permutation p -values as $\Pr\left(|\hat{\beta}_{\text{long}}^{\text{perm}}| \geq |\hat{\beta}_{\text{long}}^{\text{obs}}|\right)$ over $B = 100$ draws.

The permutation exercise is primarily informative for industries that exhibit statistically meaningful long-run effects in the baseline estimates. In particular, *Transportation and warehousing* and *Professional, scientific and technical services* display negative long-run coefficients in the baseline results and remain highly unlikely under the within-date reassignment of $\Delta \log W$ (permutation p -values equal to 0.00, i.e., below $1/B$). This pattern strengthens the interpretation that the medium-run cumulative effects in these industries are not artifacts of incidental time-series variation or sampling noise.

For *Accommodation and food services*, the baseline long-run coefficient is statistically significant under conventional clustered inference, while the permutation p -value is 0.06. This indicates that the estimated effect is still relatively unusual under random reassignment, but the evidence is weaker than for the industries above and should be interpreted with appropriate caution.

Although *Agriculture* does not exhibit a statistically significant long-run effect under conventional inference, its estimated cumulative coefficient is comparatively large in magnitude. The permutation p -value of 0.08 suggests that such a large estimate is somewhat uncommon under the null generated by within-date shuffling, motivating a cautious interpretation that the agricultural long-run estimate may reflect genuine medium-run adjustment but is not as tightly identified as the strongest cases.

For industries in which the medium-run cumulative effect is not statistically distinguishable from zero in the baseline specification, we do not interpret smaller permutation p -values as evidence of economically meaningful policy effects. In these cases, the observed estimates are themselves close to zero, and consequently provide little separation from the reference distribution

generated by random reassignments of $\Delta \log W$ across provinces.

Under such circumstances, it is mechanically possible for the permuted coefficients to exceed the magnitude of the observed estimate with relatively high frequency. This pattern does not contradict the baseline findings, but rather reflects the limited signal-to-noise ratio of the original estimates. Accordingly, permutation inference in this context is informative primarily for industries that already exhibit sizable or statistically meaningful medium-run effects in the baseline results, for which the observed coefficients can be meaningfully evaluated against the randomized benchmark.

Taken together, permutation inference provides additional support for the main conclusion that medium-run cumulative exposure matters in a subset of industries, while reinforcing the need to avoid over-interpreting scattered significance in industries without baseline evidence of long-run effects.

Table 12: Permutation Inference for Long-Run Employment Effects

Industry	Observed β_{long}	Permutation p -value	Effective B	n_{obs}
Transportation and warehousing	-0.4070	0.0000	100	4209
Services-producing sector	-0.0474	0.0000	100	4209
Educational services	-0.1007	0.0000	100	4209
Professional, scientific and technical services	-0.1566	0.0000	100	4209
Other services (except public administration)	0.1413	0.0300	100	4209
Accommodation and food services	-0.1204	0.0600	100	4209
Agriculture	0.2667	0.0800	100	4209
Wholesale and retail trade	0.0454	0.1400	100	4209
Public administration	0.0331	0.1500	100	4209
Manufacturing	-0.0486	0.1900	100	4209
Finance, insurance, real estate, rental and leasing	-0.0638	0.2600	100	4209
Forestry, fishing, mining, quarrying, oil and gas	0.2173	0.3000	100	4209
Health care and social assistance	-0.0270	0.3200	100	4209
Total, all industries	-0.0155	0.3900	100	4209
Goods-producing sector	-0.0244	0.5400	100	4209
Business, building and other support services	0.0525	0.5700	100	4209
Utilities	0.0308	0.7200	100	4125
Information, culture and recreation	0.0103	0.9000	100	4209
Construction	0.0056	0.9400	100	4209

Notes: This table reports the observed long-run effect β_{long} and the permutation p -value, which is calculated by comparing the observed effect to the distribution of effects obtained by permuting $d\log W$ within each date. The effective number of permutations (B) indicates how many valid permutations were used in the inference procedure. n_{obs} refers to the number of observations used in the regression. The permutation test helps assess the robustness of the long-run employment effect, with lower p -values indicating stronger evidence against the null hypothesis of no effect. *Source:* Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

8.6 Robustness: Spillover Effects

At first glance, regressions that include neighboring-province minimum wage changes suggest the presence of spillover effects in a small number of industries. In particular, several sectors display statistically significant coefficients on the spillover term when included alongside own-province minimum wage changes.

However, closer inspection reveals that these apparent spillover effects are unlikely to reflect identifiable causal externalities. First, own-province and spillover minimum wage changes are highly correlated across all industries, with correlation coefficients close to 0.6. This strong co-movement reflects the fact that minimum wage adjustments across provinces occur contemporaneously and are driven by common macroeconomic conditions, such as inflation and coordinated policy responses. As a result, separating own and spillover effects is subject to severe multicollinearity, limiting causal interpretation.¹¹

Second, when spillover exposure is decomposed into neighboring-province and non-neighboring (far) province components, the results do not support a geographic spillover interpretation. In several industries, spillover effects from distant provinces are as large as, or larger than, those from neighboring provinces. In other cases, neither near nor far spillovers are statistically significant. This pattern is inconsistent with mechanisms based on local labor mobility, cross-border commuting, or regional competition, which would predict stronger effects from geographically proximate provinces.¹²

Taken together, these findings indicate that the spillover coefficients primarily capture broad, economy-wide movements in minimum wages rather than true cross-province policy externalities. Consequently, although spillover terms may appear statistically significant in some specifications, they are best interpreted as reflecting common macroeconomic trends rather than identifiable spillover effects. For this reason, we conclude that minimum wage spillovers across provinces are

¹¹Minimum wage changes across Canadian provinces are highly synchronized, leading to strong correlation between own-province and spillover exposure. As a result, estimated spillover coefficients primarily capture common policy trends rather than identifiable cross-province causal effects.

¹²Even in the absence of geographic spillovers, nationally integrated labor markets or coordinated policy responses may generate correlated employment movements across provinces.

not separately identifiable in the present data.

Table 13: Own and Neighboring-Province Minimum Wage Effects on Employment

Industry	Own $\Delta \log W$	p -value	Neighbor Spillover	p -value
Transportation and warehousing	−0.271***	0.008	0.370***	0.000
Forestry, mining, oil and gas	0.239	0.310	0.843***	0.002
Professional, scientific and technical services	−0.239**	0.028	0.261*	0.083
Business support services	0.152	0.218	−0.300*	0.093
Accommodation and food services	−0.064	0.419	0.166	0.240
Manufacturing	−0.039	0.653	0.024	0.845
Wholesale and retail trade	0.044*	0.089	0.014	0.797
Total, all industries	−0.002	0.961	0.049	0.337

Notes: Each row reports estimates from a two-way fixed effects regression including province and date fixed effects. Standard errors are clustered at the province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Source: Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

Table 14: Correlation Between Own and Spillover Minimum Wage Changes

Industry	Corr($\Delta \log W$, Spillover $\Delta \log W$)
Accommodation and food services	0.591
Manufacturing	0.591
Professional services	0.591
Wholesale and retail trade	0.591
Health care and social assistance	0.591
Goods-producing sector	0.591
Services-producing sector	0.591
Total, all industries	0.591
Utilities	0.580

Notes: Correlations are computed at the province–date level within each industry. The uniformly high correlation reflects strong contemporaneous co-movement of minimum wage changes across provinces.

Source: Author's calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

Table 15: Near vs. Far Province Spillover Effects

Industry	Near Spillover	Far Spillover	Interpretation
Manufacturing	−0.369*	−1.847**	Far > Near
Information, culture and recreation	−0.897***	−4.345***	Far > Near
Services-producing sector	0.130***	0.367	Near only
Forestry, mining, oil and gas	−0.235	−5.076***	Far only
Transportation and warehousing	0.272	−0.459	Neither
Accommodation and food services	0.129	−0.171	Neither
Total, all industries	0.098	0.234	Neither

Notes: Each regression includes own $\Delta \log W$, near-province spillover, far-province spillover, and province and date fixed effects. Standard errors are clustered at the province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *Source:* Author’s calculations using Statistics Canada and Employment and Social Development Canada (ESDC) data.

9 Conclusion

Motivated by standard theoretical frameworks emphasizing labor-market frictions, adjustment costs, and potentially nonlinear policy effects, this paper studies the employment consequences of frequent provincial minimum wage adjustments in post-pandemic Canada. Using a newly constructed province–industry–month panel and an industry-specific stacked event-study design with dose-based treatment intensity, we estimate dynamic employment responses in a quasi-experimental setting characterized by staggered and repeated policy changes.

Three main conclusions emerge. First, minimum wage increases do not generate broad-based short-run employment losses across industries. After accounting for small-sample inference using province-level wild cluster bootstrap procedures, joint tests of short-run event-time coefficients are generally not statistically significant. This suggests that much of the apparent short-run significance obtained under conventional cluster-robust inference reflects over-rejection in a setting with a limited number of policy clusters, rather than robust evidence of immediate employment effects.

Second, evidence of medium-run employment adjustments is concentrated in a single industry. Specifically, accommodation and food services exhibits a statistically significant negative cumulative employment response to sustained minimum wage increases that remains robust under wild cluster bootstrap inference. In contrast, estimated medium-run effects in other industries are im-

precisely estimated and cannot be distinguished from zero once inference is adjusted for the small number of clusters.

This finding differs from influential early evidence based on the fast-food sector, such as the New Jersey–Pennsylvania natural experiment studied by Card and Krueger (1994), which documented little to no employment response to a single minimum wage increase. A key distinction in the Canadian context is the combination of a relatively high minimum wage level and the high frequency of policy adjustments in the post-pandemic period. Rather than a one-time discrete change from a low baseline, provinces implemented repeated increases that cumulatively raised the wage floor to historically elevated levels. These results therefore suggest that medium-run employment responses are not widespread across the economy, but may emerge in sectors with high exposure to minimum wage policies once adjustments accumulate from an already elevated baseline. While the analysis does not identify a precise threshold, the findings can be interpreted as a cautious signal that frequent minimum wage increases may begin to generate detectable employment adjustments in the most exposed low-wage industries.

Third, the main conclusions are supported by a broad set of robustness exercises. The results are stable across alternative event-window definitions, placebo timing tests, permutation-based inference, and specifications allowing for potential cross-province spillovers. In particular, apparent spillover patterns are more consistent with strong contemporaneous co-movement in provincial policy changes than with identifiable cross-province causal externalities. Taken together, these diagnostics reinforce the interpretation that the estimated employment responses are not mechanically driven by macroeconomic synchronization, event timing choices, or specific modeling assumptions.

Several limitations should be noted. The analysis focuses on the post-pandemic period and lacks a long, stable pre-pandemic baseline. Employment outcomes during the recovery phase may exhibit elevated volatility and structural change, and the estimated medium-run effects should therefore not be interpreted as long-run structural impacts. Future research using longer time horizons and additional margins of adjustment, such as hours worked, earnings distributions, firm entry

and exit, and non-wage benefits, would be valuable for assessing whether the industry-specific adjustments identified here persist, attenuate, or reverse over time.

Overall, this paper finds that recent high-frequency minimum wage increases in Canada did not produce widespread immediate employment losses. However, sustained increases are associated with cumulative medium-run employment declines in accommodation and food services, a sector characterized by high minimum wage exposure and limited short-run adjustment capacity. The contribution of the paper is primarily empirical and causal with respect to employment dynamics, providing evidence on the heterogeneous and time-dependent nature of minimum wage effects that can inform subsequent welfare-based evaluations of minimum wage policy.

References

- Autor, D., Dorn, D., Hanson, G., and Majlesi, K. (2019). Importing political polarization? the electoral consequences of rising trade exposure. *American Economic Review*, 109(10):3139–3183.
- Bhaskar, V. and To, T. (1999). Minimum wages for ronald mcdonald monopsonies: A theory of monopsonistic competition. *Economic Journal*, 109(455):190–203.
- Brochu, P. and Green, D. A. (2013). The impact of minimum wages on labour market transitions. *Economic Journal*, 123(573):1203–1235.
- Cabral, L. (2017). *Introduction to Industrial Organization*. MIT Press, Cambridge, MA.
- Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.
- Card, D. and Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in new jersey and pennsylvania. *American Economic Review*, 84(4):772–793.
- Clemens, J. and Strain, M. R. (2017). Minimum wage employment effects: New evidence from monthly data. *Journal of Public Economics*, 150:1–18.
- Hamermesh, D. S. (1993). *Labor Demand*. Princeton University Press, Princeton, NJ.
- MacKinnon, J. G. and Webb, M. D. (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254.
- Manning, A. (2003). *Monopsony in Motion: Imperfect Competition in Labor Markets*. Princeton University Press, Princeton, NJ.
- Manning, A. (2021). *Monopsony in Labor Markets: A Review*. IZA World of Labor.
- Mian, A. and Sufi, A. (2014). What explains the 2007–2009 drop in employment? *Econometrica*, 82(6):2197–2223.

Stigler, G. J. (1946). The economics of minimum wage legislation. *American Economic Review*, 36(3):358–365.

A Wild Cluster Bootstrap Inference

This appendix reports inference based on wild cluster bootstrap procedures designed for difference-in-differences settings with a small number of clusters. Because minimum wage policies in Canada are implemented at the provincial level, the effective number of clusters in the analysis is limited to ten provinces. In such settings, conventional cluster-robust standard errors may suffer from finite-sample bias and over-reject the null hypothesis.

A.1 Bootstrap Procedure

To address this concern, we implement a province-level wild cluster bootstrap using Rademacher weights, following the methodology proposed by Cameron et al. (2008) and MacKinnon and Webb (2017). Inference is conducted separately for each industry-specific stacked difference-in-differences regression.

For each hypothesis test, we apply a *restricted* wild bootstrap procedure. Specifically, we first estimate a restricted model imposing the null hypothesis (e.g., setting the cumulative exposure coefficient or the relevant event-time coefficients equal to zero) and obtain fitted values and residuals from this model. Bootstrap outcome variables are then constructed as

$$Y_{i,s,t}^* = \hat{Y}_{i,s,t}^0 + \hat{u}_{i,s,t}^0 \cdot w_s,$$

where $w_s \in \{-1, +1\}$ is a Rademacher weight drawn at the province level. For each bootstrap draw, the unrestricted model is re-estimated using the bootstrap outcome, and the same test statistic as in the original sample (either a t -statistic for a single coefficient or a Wald statistic for joint hypotheses) is recomputed. Bootstrap p -values are calculated as the proportion of bootstrap test statistics that exceed the original statistic in absolute value.

This procedure ensures that bootstrap inference is conducted in the same statistical space as the original test and remains valid under a small number of clusters.

A.2 Bootstrap Tests Considered

We report wild cluster bootstrap inference for two classes of hypotheses: (i) the cumulative medium-run exposure effect captured by the coefficient on $\Delta \log W_{s,t}$, and (ii) joint tests of the short-run event-time coefficients. For joint hypotheses, we separately test the null that all event-time coefficients are equal to zero (JOINT), as well as pre-treatment and post-treatment subsets.

A.3 Results

Table 16 reports wild cluster bootstrap p -values alongside conventional cluster-robust p -values for selected industries. Two main patterns emerge. First, bootstrap p -values are generally larger than their cluster-robust counterparts, indicating that conventional clustered inference tends to over-reject the null in this setting. Second, after applying the bootstrap correction, evidence of statistically significant employment responses is concentrated in a small subset of low-wage industries. In particular, the cumulative exposure effect remains statistically significant for accommodation and food services, while most short-run dynamic responses are no longer jointly significant across industries.

Overall, these results suggest that the main conclusions of the paper are not driven by spurious precision arising from a limited number of clusters, and that robust evidence of employment effects is confined to specific sectors.

Table 16: Wild Cluster Bootstrap Inference by Industry (Province-Level Clustering)

Industry	Cumulative exposure ($\Delta \log W_{s,t}$)		Joint event-time test	
	Cluster p	Wild bootstrap p	Cluster p	Wild bootstrap p
Accommodation and food services	0.009	0.047	0.000	0.033
Agriculture	0.274	0.433	0.001	0.253
Business, building and other support services	0.636	0.647	0.594	0.773
Construction	0.953	0.933	0.875	0.953
Educational services	0.162	0.953	0.029	0.227
Finance, insurance, real estate, rental and leasing	0.518	0.700	0.241	0.600
Forestry, fishing, mining, quarrying, oil and gas	0.409	0.693	0.562	0.667
Goods-producing sector	0.787	0.867	0.079	0.420
Health care and social assistance	0.241	0.660	0.000	0.193
Information, culture and recreation	0.468	0.860	0.000	0.447
Manufacturing	0.274	0.380	0.001	0.193
Other services (except public administration)	0.636	0.773	0.774	0.900
Professional, scientific and technical services	0.361	0.360	0.000	0.253
Public administration	0.347	0.147	0.000	0.147
Services-producing sector	0.184	0.447	0.000	0.547
Total, all industries	0.131	0.493	0.000	0.600
Transportation and warehousing	0.227	0.427	0.000	0.527
Utilities	0.787	0.787	0.000	0.433
Wholesale and retail trade	0.118	0.413	0.000	0.473

Notes: The table reports p -values from conventional cluster-robust inference and province-level wild cluster bootstrap inference for industry-specific stacked difference-in-differences regressions. Wild bootstrap p -values are obtained using a restricted wild cluster bootstrap with Rademacher weights and 999 replications. Minimum wage changes are clustered at the provincial level. Only accommodation and food services exhibits statistically significant medium-run employment responses after bootstrap correction.